

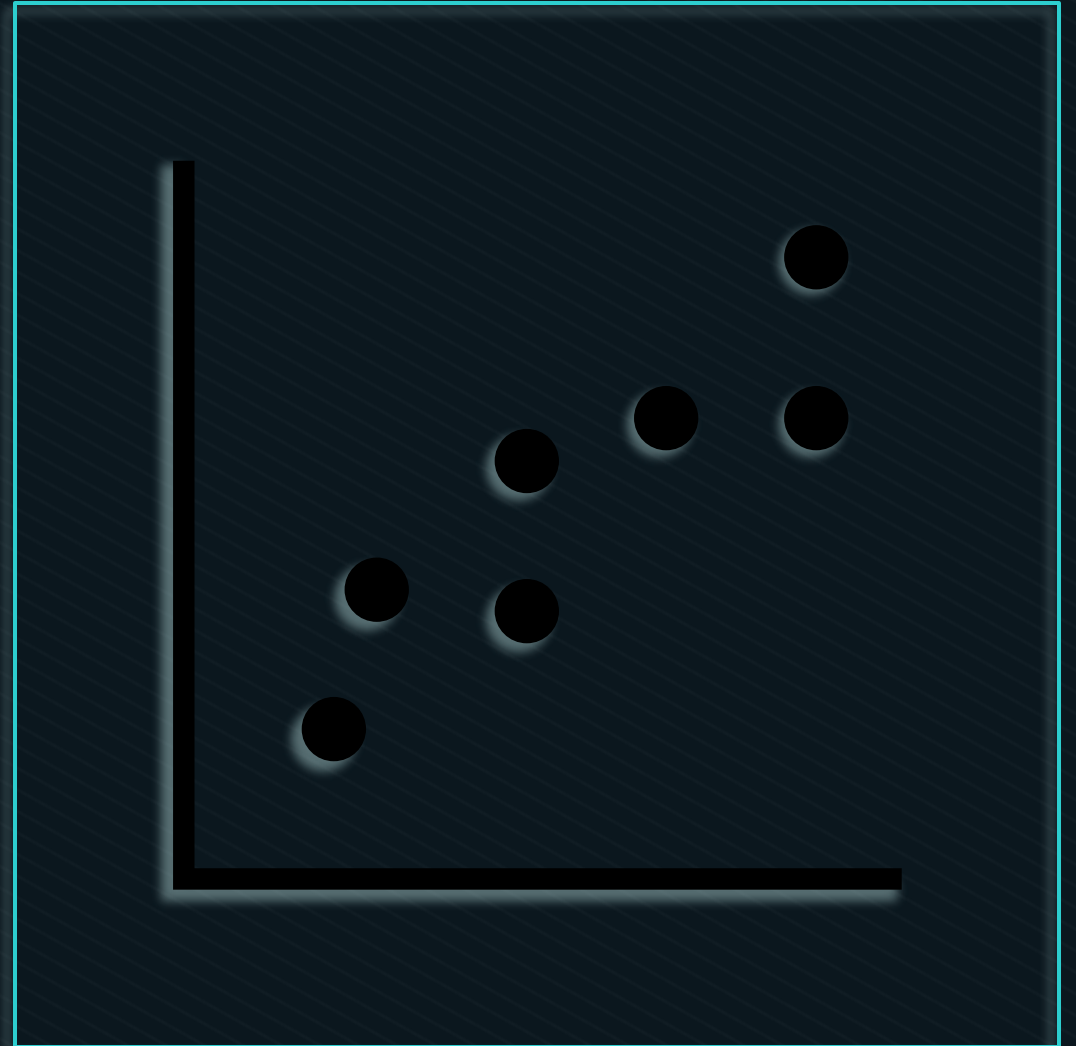
Part 1: EDA

Assumptions

- Count is the sum of casual and registered users
- Holiday and Working day are mutually exclusive, this was verified through filtering the data
- Working day = 0 is the weekend

Statistical Significance

- The first step taken was to determine the size and strength of the relationship between the variables. One cannot assume significance.
- Three different techniques were used to measure statistical significance based on the characteristics of the variable: Pearson Correlation, Correlation Ratio and Point Biserial Correlation

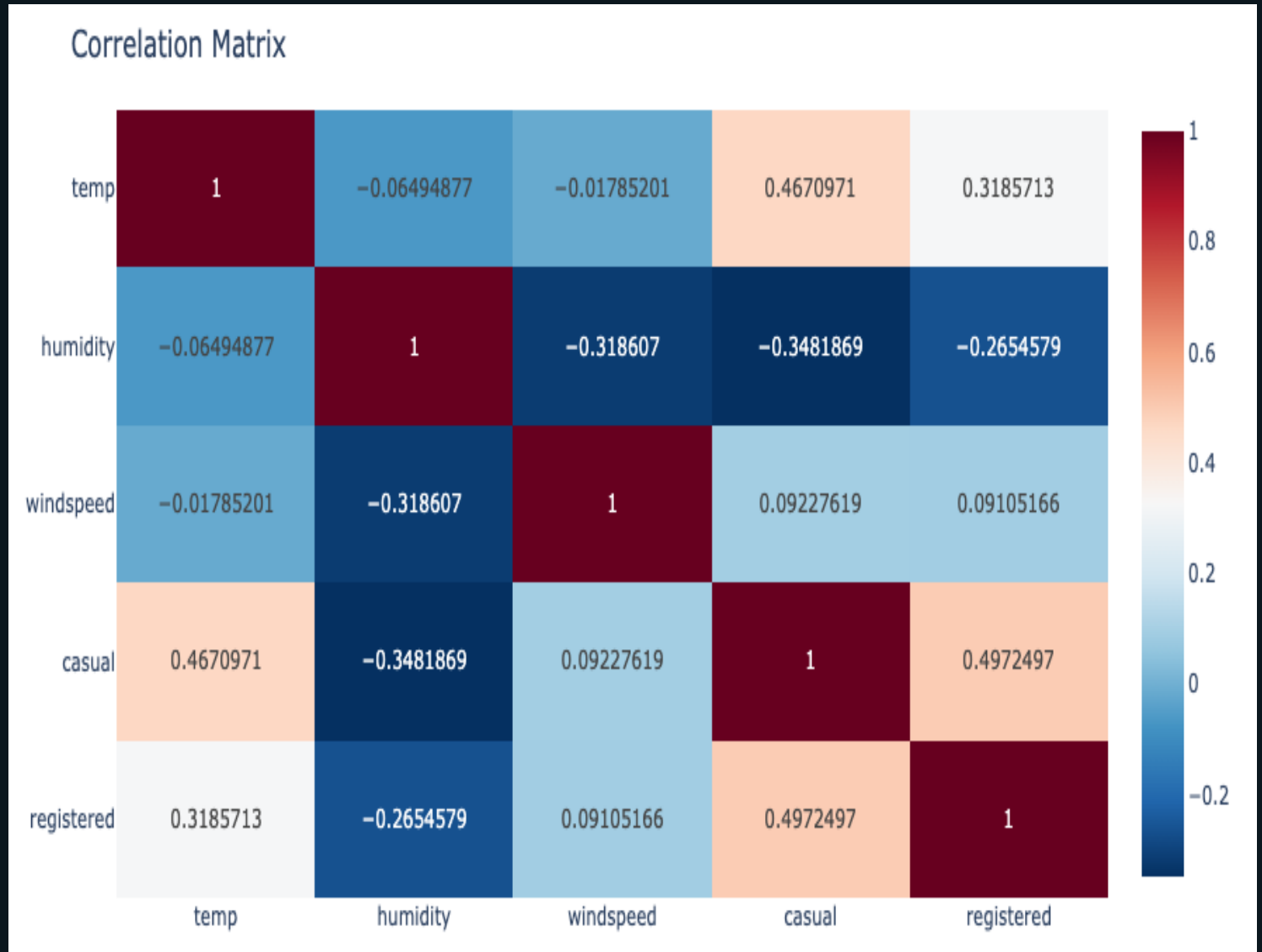


Pearson Correlation Matrix

- This measures the statistical relationship between two continuous variables assuming linearity and homoscedasticity.
- From the graph we can see that temperature and humidity have the strongest impact; temperature positive and humidity negative.

Theories:

- 1) As temperature increases casual and registered users are more likely to rent a car.
- 2) High humidity can often correlate to bad weather, perhaps people are more likely to stay indoors when the weather feels “muggy”.



Correlation Ratio

Measures the correlation between a continuous and categorical variable with more than 2 categories.

Assumptions/Caution:

- 1) Assumes asymmetry variables (depends on the variable type)
- 2) Cannot prove causal direction only level
- 3) Best used when categorical data is non-ordinal

	Season	Holiday	Workingday	Weather
Total	0.2476	0.0054	0.0116	0.1332
Casual	0.2946	0.0438	0.3191	0.1366
Registered	0.2104	0.0210	0.1195	0.1154

Season and weather seem to have the overall most significant relationship. However, it seems that working day has the overall strongest relationship to casual users. Unfortunately, we will have to investigate more to understand the casual direction of the relationship.

Point-biserial Correlation

Measures the correlation between a continuous and categorical variable with 2 categories.

Pairs	Value	Statistical Significance
Holiday and Count	-0.005	NO
Working day and Count	0.012	NO
Holiday and Casual	0.044	YES
Working day and Casual	-0.319	YES
Holiday and Registered	-0.021	YES
Working day and Registered	0.119	YES

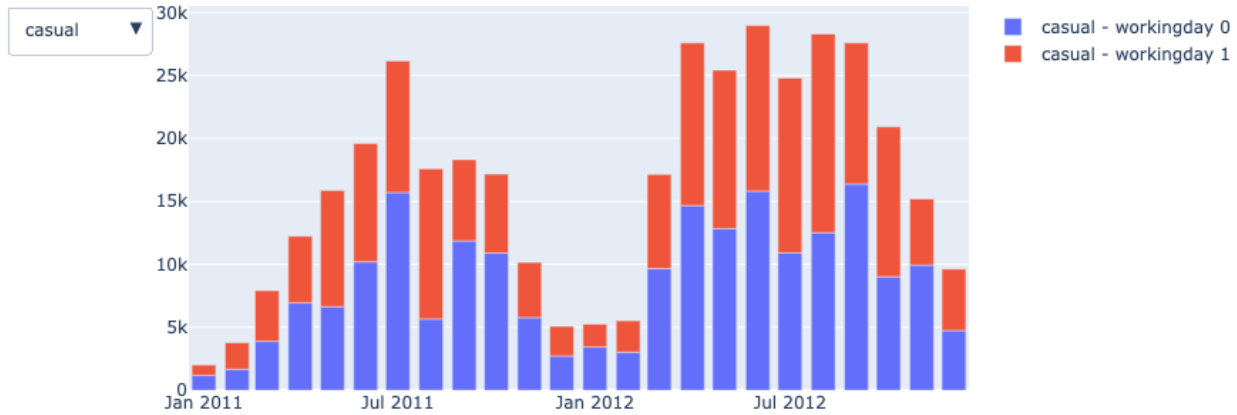
These results more or less confirm the results from the correlation ration except this time we can see the direction of the relationship. We see that holidays have a positive and weak relationship with casual users but a negative one for registered users.

Working days have the strongest impact; negative for casual users but positive for registered users.

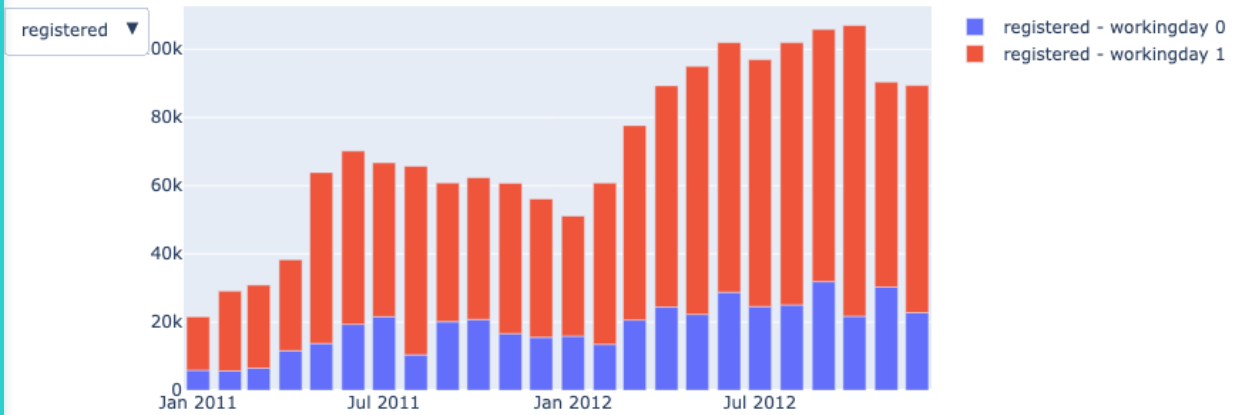
Conclusion: User type greatly affects the relationship to the other variables, they should be analysed separately.

Seasonality

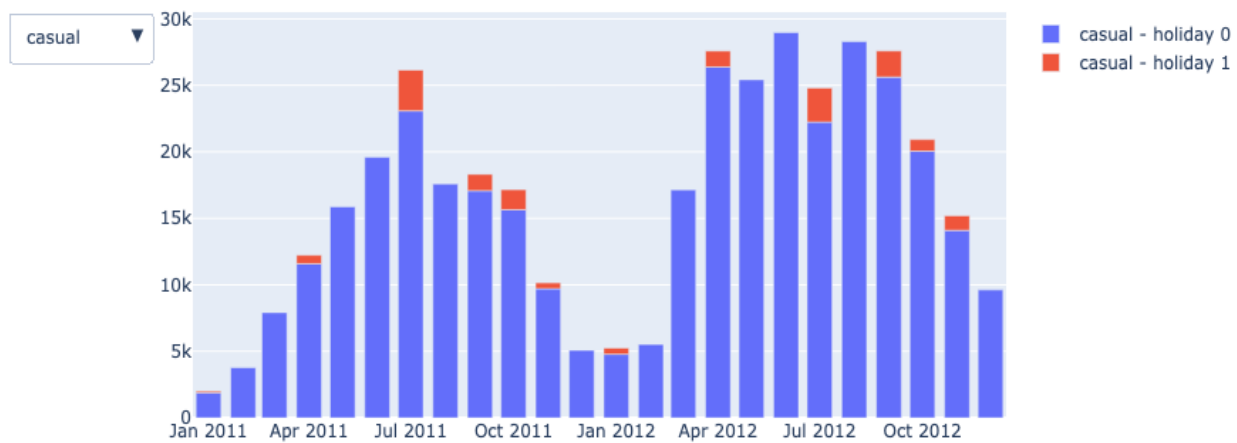
casual by Month



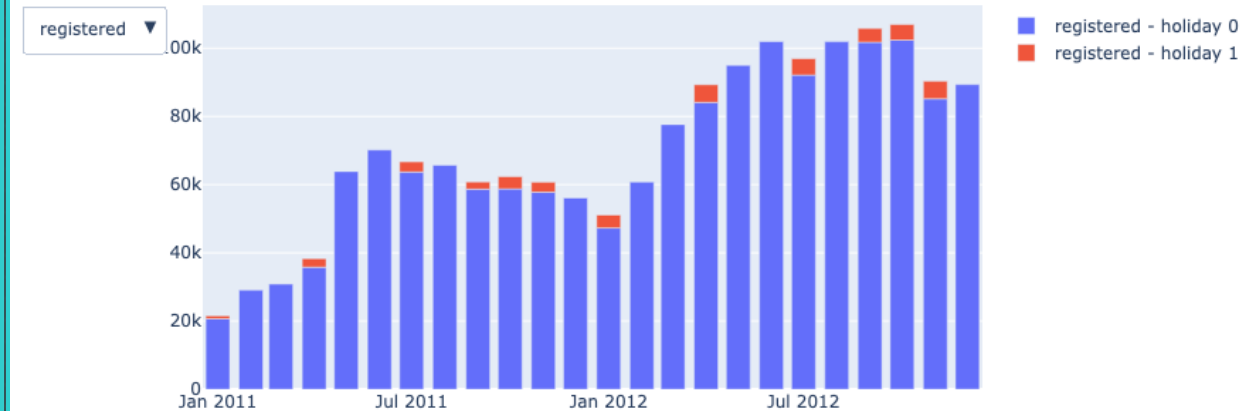
registered by Month



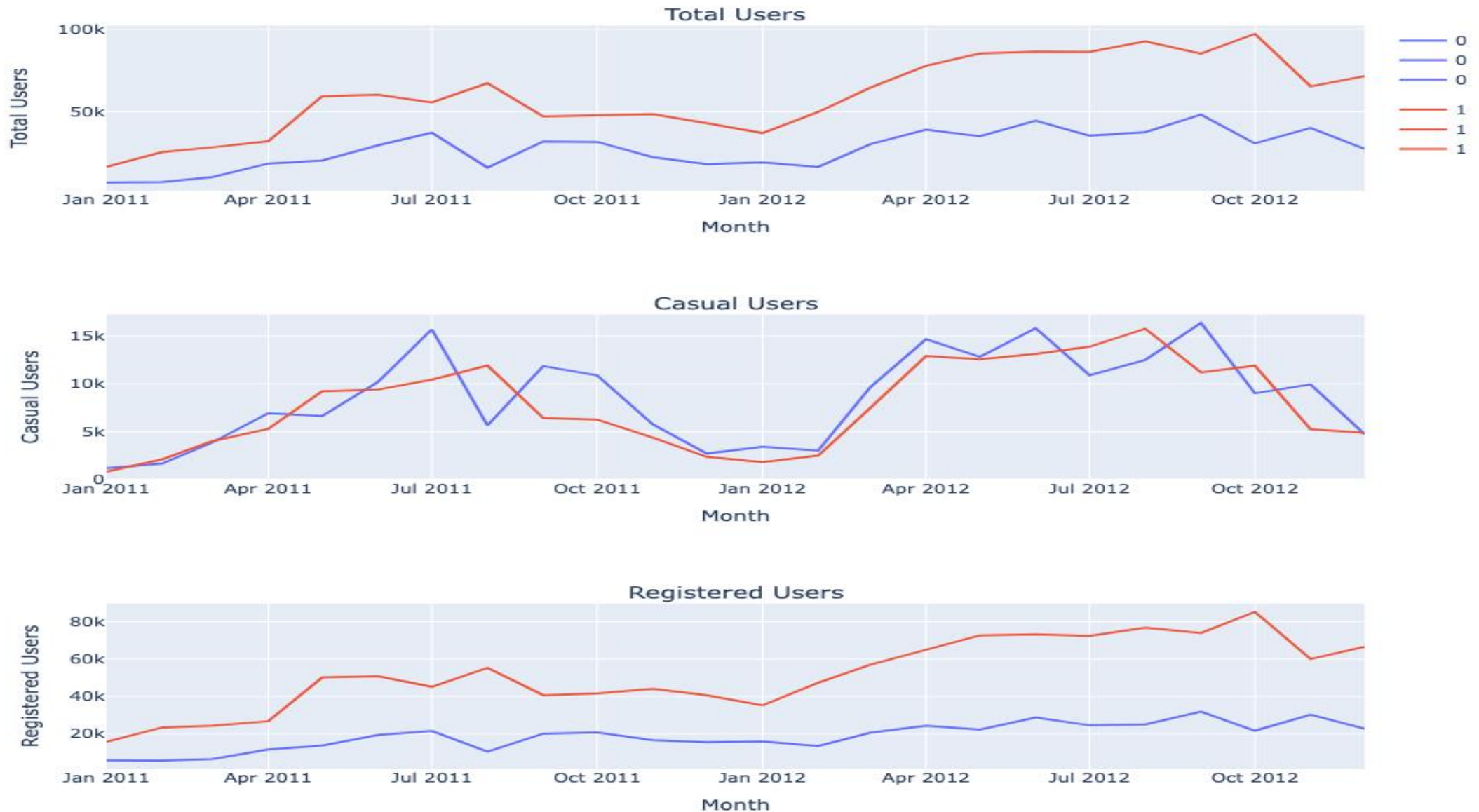
casual by Month



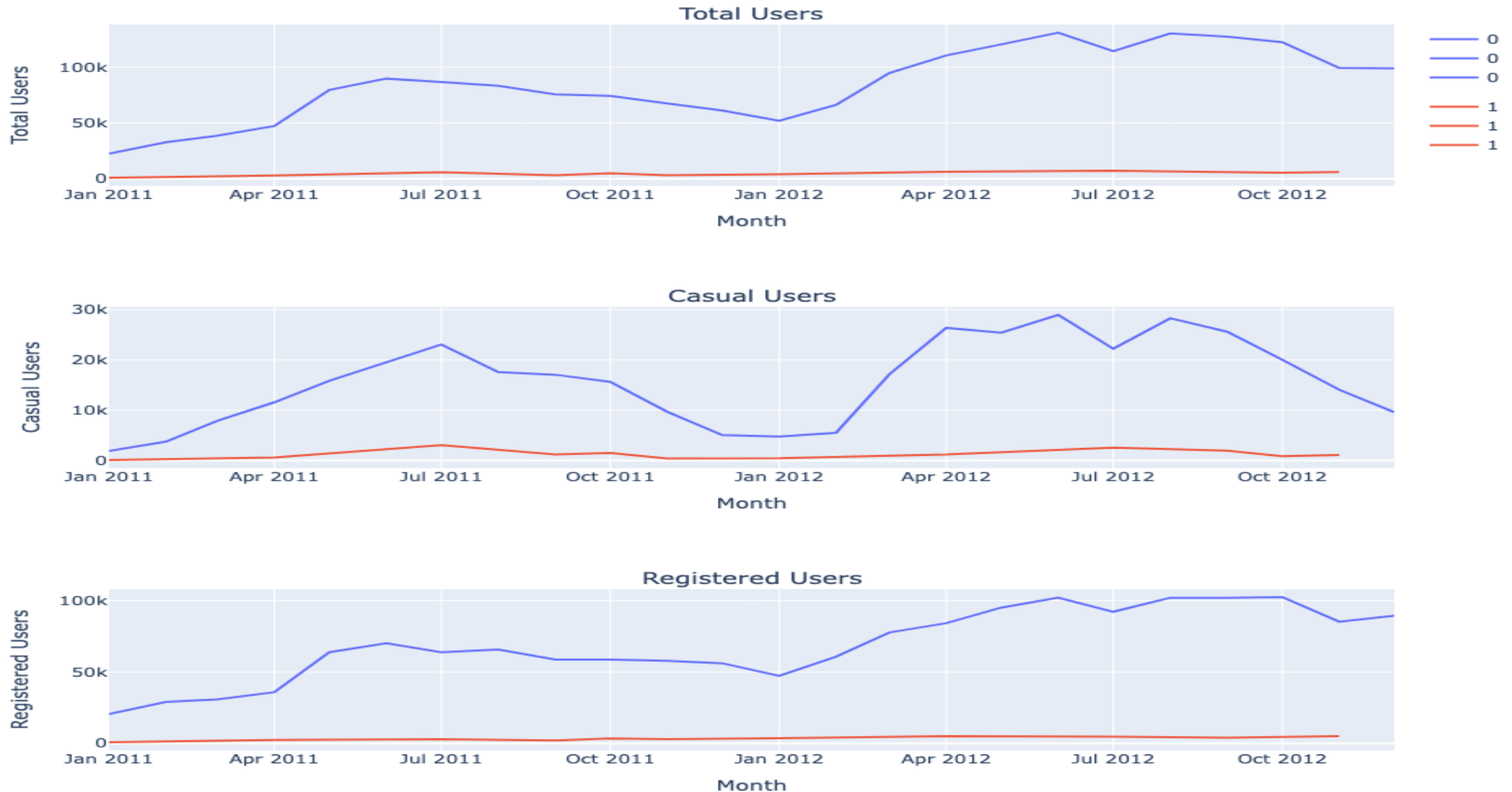
registered by Month



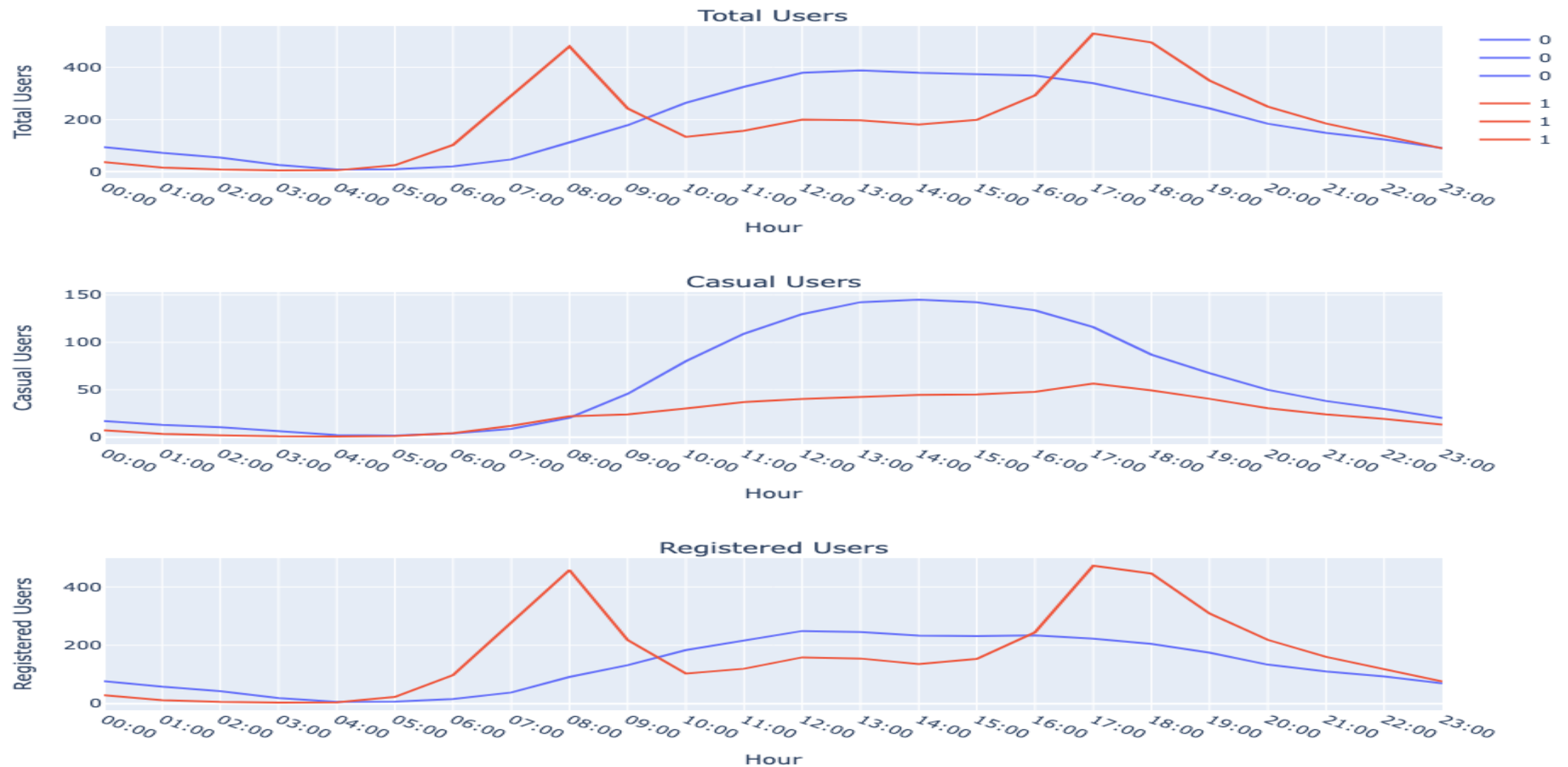
Users by Month and Working day



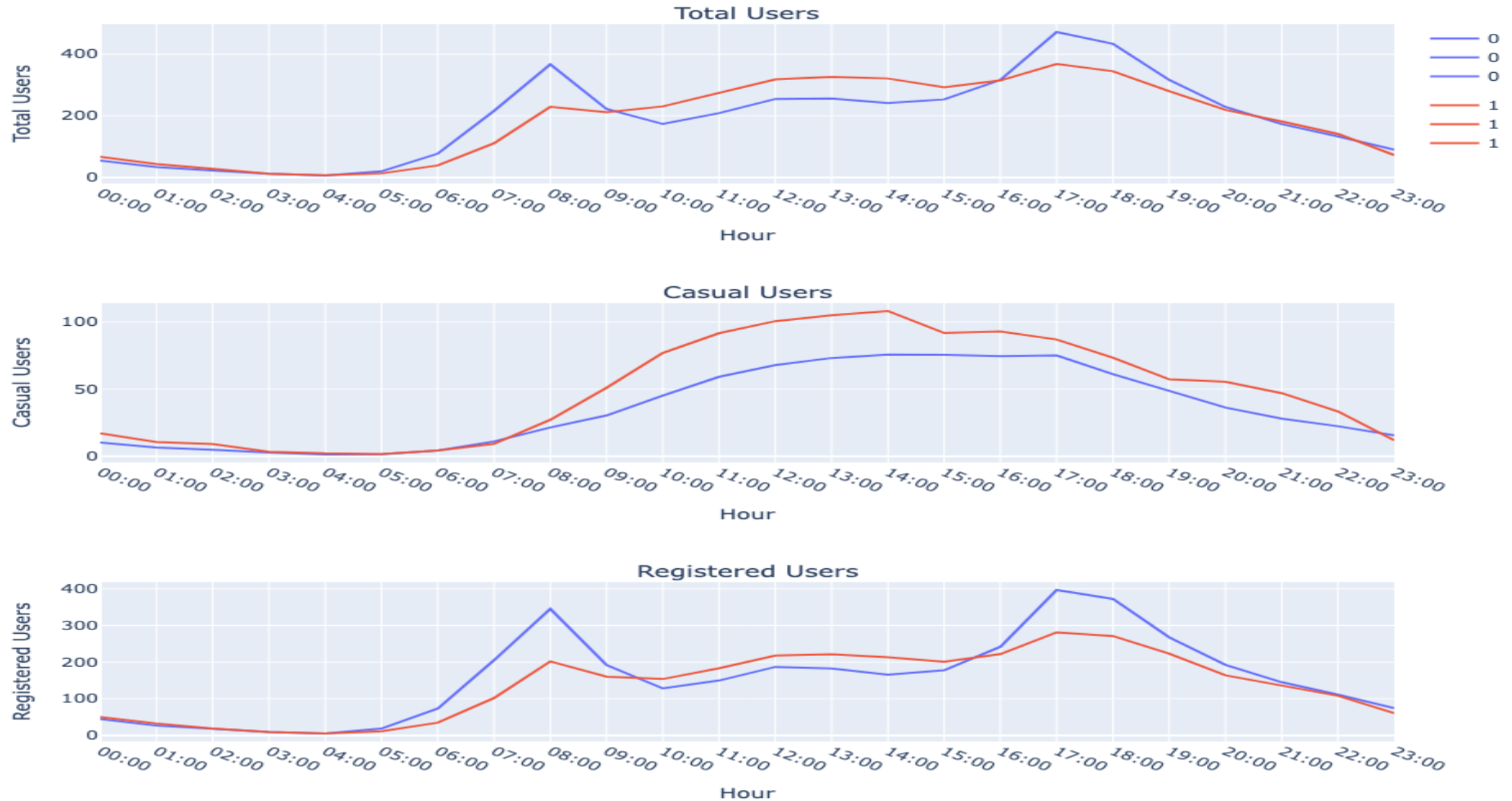
Users by Month and Holiday



Users by Hour and Working day



Users by Hour and Holiday



Interactive Graph: Temperature

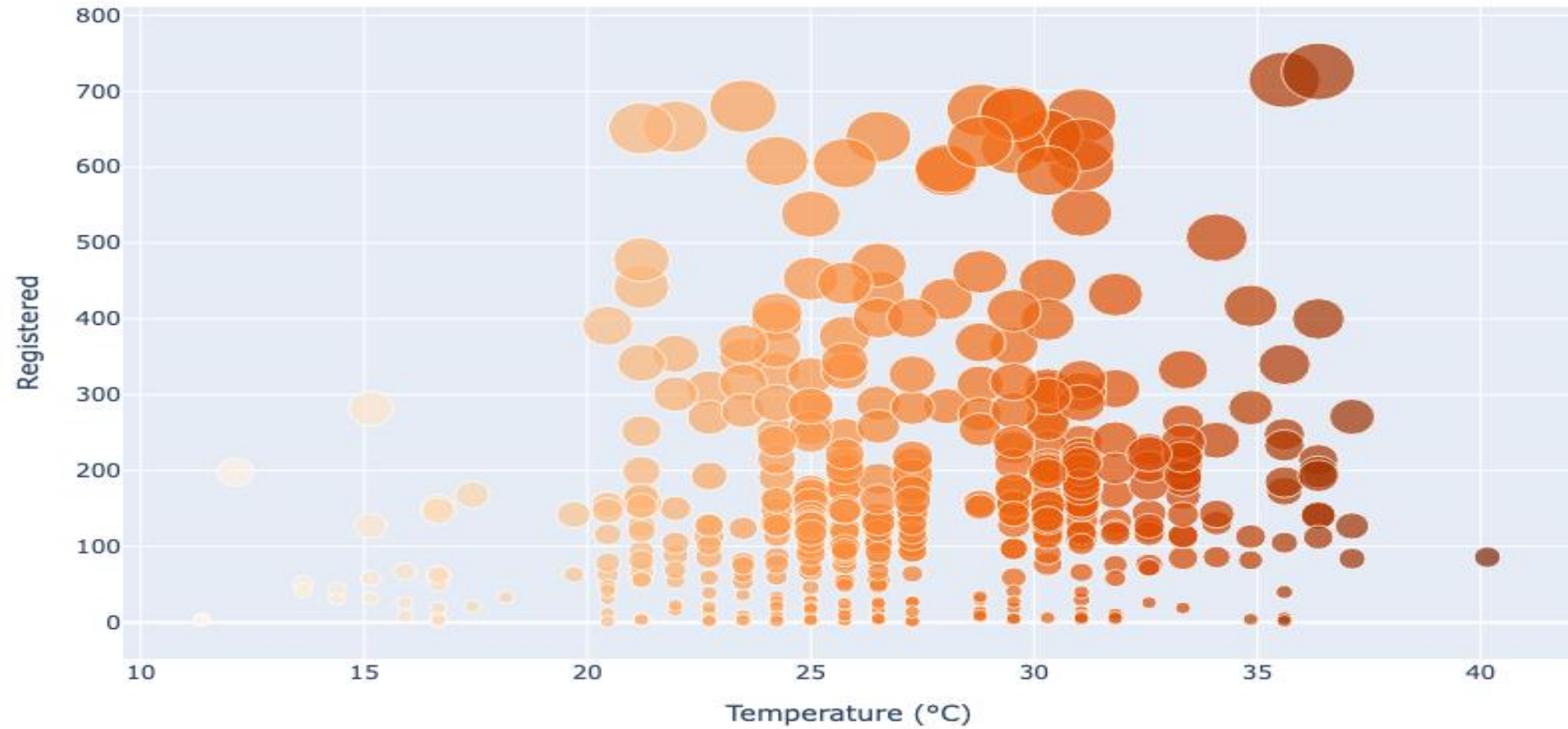
Temperature and Registered (Spring, Weekday, Cle

Spring

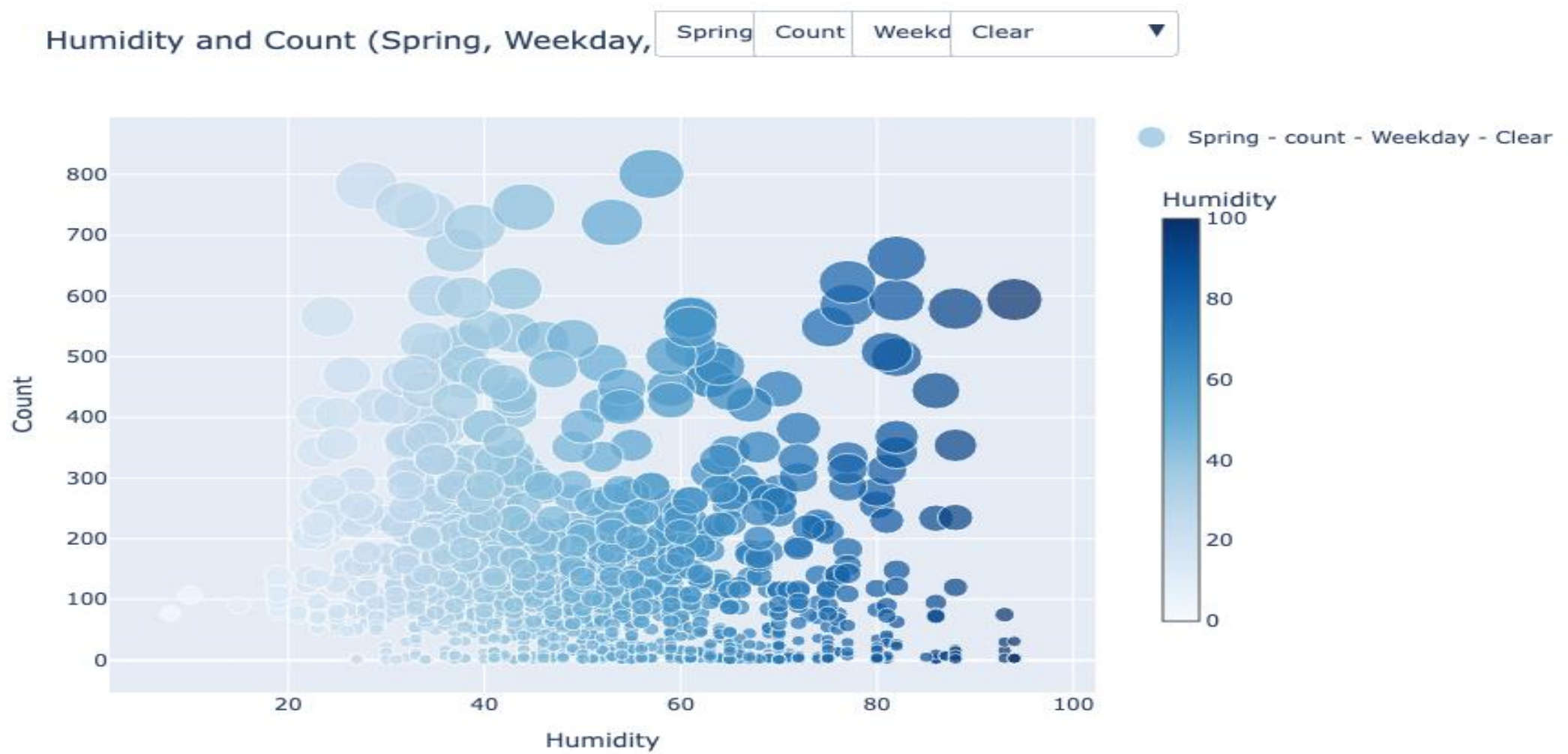
Registere

Weekend

Clear



Interactive Graph: Humidity



Part 2: Simulation

Assumptions:

- 1) The first appearance of a car is at 100% battery life minus the km driven in that row.
- 2) Probabilities given are in percentages, so dividing by 100 gives the raw probability.
- 3) Rental duration is Start time subtracted from End time.
- 4) Revenue per minute is Revenue divided by Revenue collected from customer.
- 5) Car battery has 100% efficiency.
- 6) If battery consumption is 0.5% per km, then a car with a battery of 100% efficiency can drive for 200 km.
- 7) If the trip has lasted longer than 200km then the customer charged the car.
- 8) Data has been sorted in chronological order to enable assignment of car id variable.
- 9) A car first appears in the dataset with a rental state of GREEN.
- 10) Car state is switched to RED when battery falls below 20%, i.e. after 160km.
- 11) If one row has a distance that exceeds 160km, we assume that the customer took the car in to charge.
- 12) Customer always parks their car at a charging station at the end of a trip when in RED state.
- 13) Car recharge time is 6 hours. Minimum time to get back on road = $180 + (6 * 60) = 540$ minutes
- 14) There is a delay of MINIMUM 180 minutes + 6 hours recharge time before a car in RED state is allowed back on the road i.e. GREEN
- 15) No car overlap. If the end time for GREEN state is greater than the next start time, then this is a different car with a different id.
- 16) No maintenance is done other than charging the car for the duration of the dataset.
- 17) If State is RED but Revenue and Distance are 0, then we assume is this a cancelled ride.
- 18) If the State is GREEN but no Revenue, then this was a car put up for rent but left unchosen.

Method:

1. Dataset was sorted in chronological order using the Start time
2. A new variable called car_id was introduced that respected the assumptions listed
3. Battery remaining was calculated
4. Filter out cars in GREEN state as we are only interested in revenue generating vehicles
5. Data was segmented into sessions. A session is the entire period a car is on the road until it is recharged again
6. Do calculations listed on slide 18
7. Remove all rows that do not have profitable minutes
8. Apply decision criteria
9. As there is no formula given for revenue, only that it is slightly negatively correlated with distance, we will maximise profitable minutes instead of revenue as it is more consistent.
10. Create an app to display results

Decision criteria:

Session must have accumulated profitable minutes while in RED state. Session must be last session right before recharge as this is the most profit maximising (maximum profitable minutes). The most maximising ride is always the last ride before charging. Results given in profitable_rides.csv

Formulas:

Battery remaining=
 $100 - ((\text{accumulated distance} / 200) * 100)$

Incentive cost =
 $\text{Incentive} * 0.29$

Breakeven point =
 $\text{Recharge minutes} + \text{Charging time}$

Incentive
Saving/Loss =
 $\text{Charging Cost} - \text{Incentive Cost}$

Running Duration =
Cumulative ride
duration per session

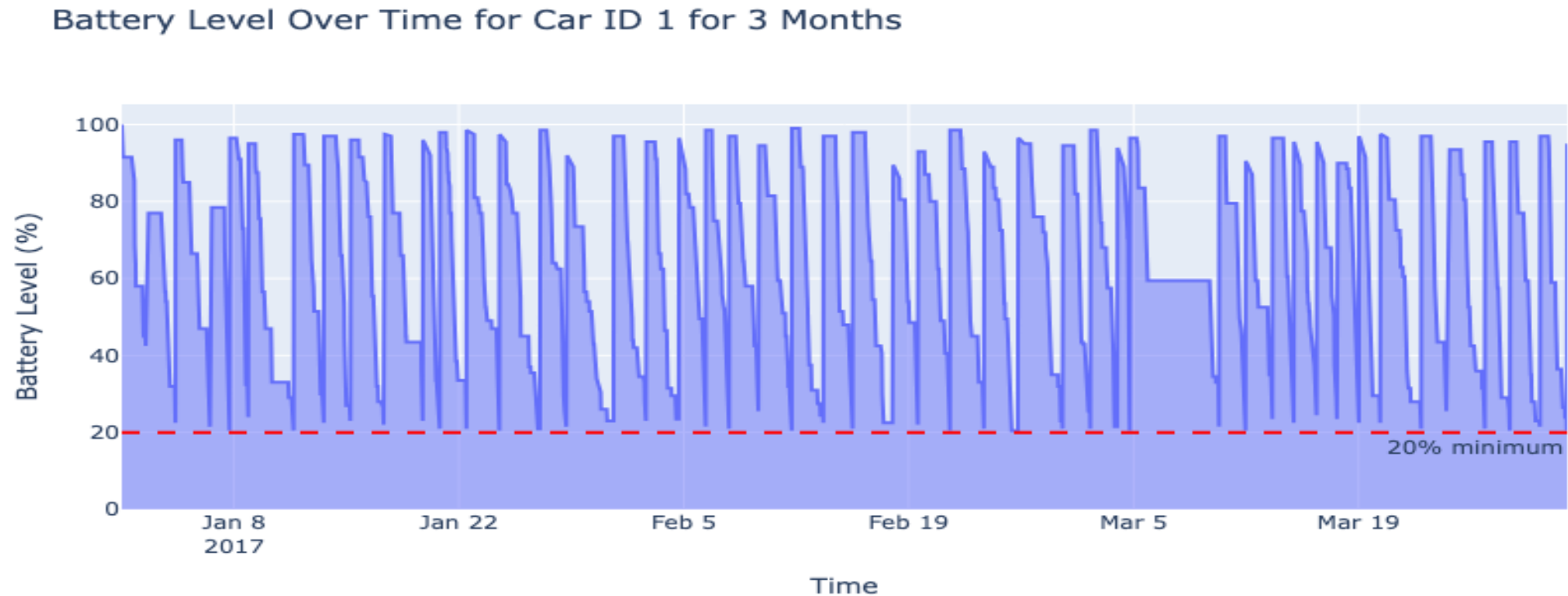
Running Revenue =
Cumulative revenue
per vehicle session

Profitable Minutes =
 $\text{Running Duration} - \text{Breakeven point}$

Incentive 20, 15, 10,
5 = Profitable
minutes – Incentive
Minutes

Battery Level Over Time for Car ID 1 for 3 Months

- Raw data can be found in car_id_1.csv in the RESULTS folder.



Incentives

	Incentive	Charging_probability	Probability_Percent	Incentive_Cost	Incentive_Savings/Loss
0	0	0	0.00	0.00	6.00
1	5	8	0.08	1.45	4.55
2	10	16	0.16	2.90	3.10
3	15	24	0.24	4.35	1.65
4	20	32	0.32	5.80	0.20
5	25	40	0.40	7.25	-1.25
6	30	48	0.48	8.70	-2.70
7	35	56	0.56	10.15	-4.15
8	40	64	0.64	11.60	-5.60
9	45	72	0.72	13.05	-7.05
10	50	80	0.80	14.50	-8.50

Based on these calculations we can conclude that an incentive greater than 20 minutes creates a loss with respect to charging costs.

Decision App

Ride Incentives Suggestions

Select Car ID: 2 ▼

Select Start Date: 2017-01-13 × ▼

Start	End	Revenue	Distance	Ride Duration (Minutes)	Car ID	Remaining Battery	Profitable Minutes	Free 20mins	Free 15mins	Free 10mins	Free 5mins
filter date											
2017-01-13	2017-01-13	2	3	11	2	47	8	-12	-7	-2	3
2017-01-13	2017-01-13	2	4	12	2	45	20	0	5	10	15
2017-01-13	2017-01-13	4	6	14	2	42	34	14	19	24	29
2017-01-13	2017-01-13	2	2	6	2	41	41	21	26	31	36

User can filter car and date to see at which level they would like to reward the user.

Alternative Method

Model Suggestions

Gradient Descent

- Algorithm to maximise profitable minutes

Hyperparameter Tuning

Bayesian Optimisation

- This method allows us to optimize based on constraints such as battery level and adjust based on demand

Applications

- We can calculate the battery level threshold that results in the maximum amount of profitable minutes
- Although this method is not the fastest it does not need to be. The thresholds once calculated can be applied as rules in the app.
- Monitoring pipelines can also be added to ensure the model remains within bounds
- Great for edge cases, e.g when battery is low that people don't want to rent for large trips but not so low it can't do short trips.

